Midterm Review Quiz

50 minutes • 50 points • Through continuous distributions

Name: _____ **Date:** _____

Instructions. Answer all five problems and show the essential reasoning. No calculator is needed. Exact expressions, including factorials, exponentials, and values of the standard normal CDF Φ, are acceptable. You may quote standard distribution and moment formulas. Geometric variables count trials *including the first success*; exponential parameters are *rates*. Reserve the final 5 minutes for checking.

1. A random permutation

10 points · about 9 minutes

Four travelers' luggage tags are shuffled uniformly among all 4! assignments, one tag per traveler. Let I_i indicate that traveler i gets the correct tag, and let $X = I_1 + I_2 + I_3 + I_4$.

(a) Find $\mathbb{P}(X = 0)$ using inclusion–exclusion. Show your count or sum. (4 points)

(b) Find $\mathbb{E}[X]$ by using the indicators. Does this calculation require the indicators to be independent? (2 points)

(c) For $i \neq j$, find $\mathbb{P}(I_i = I_j = 1)$ and decide whether I_i, I_j are independent. Then compute $\mathrm{Var}(X)$, keeping the cross terms. (4 points)

2. One hidden choice, two observations

10 points · about 8 minutes

A machine is selected once: machine A with probability $1/3$ and machine B with probability $2/3$. The selected machine produces two items. Conditional on the selected machine, defect indicators for the items are independent. Each item from A is defective with probability $1/4$; each from B is defective with probability $1/8$. Let D_1, D_2 be the events that the first and second items are defective.

(a) Find $\mathbb{P}(D_1)$. (3 points)

(b) Given that the first item is defective, find the probability that machine A was selected. (3 points)

(c) Find $\mathbb{P}(D_2 \mid D_1)$. Are D_1, D_2 independent without conditioning on the machine? Justify your answer numerically. (4 points)

3. Counts, waits, and a complete collection

10 points · about 9 minutes

Each pack contains one of three sticker types, independently and uniformly. Packs may duplicate previously seen types.

(a) Let N count type-1 stickers in the first four packs. State its distribution and parameters, and find $\mathbb{P}(N = 2)$ and $\mathbb{E}[N]$. (3 points)

(b) Exactly two distinct types have been collected. Let G be the number of *additional* packs through the first pack containing the missing type. State its distribution and parameter, and find $\mathbb{E}[G]$ and $\mathbb{P}(G > 2)$. (3 points)

(c) Starting with no stickers, let T be the number of packs through the first complete collection. Split T into the waits for the first, second, and third distinct types, and compute $\mathbb{E}[T]$. Explain why T itself is not one geometric random variable. (4 points)

4. Choose the law and use its parameters

10 points · about 9 minutes

(a) Calls arrive with Poisson counts: an interval of length t hours contains Poisson$(3t)$ calls, and disjoint intervals have independent counts. Let M count calls in the next 20 minutes. State its distribution and parameter, and find $\mathbb{P}(M \geq 2)$. (3 points)

(b) For the same call model, let W be the time in hours until the next call. Use the probability of zero calls to find $\mathbb{P}(W > t)$ for $t \geq 0$. Identify the distribution of W and find $\mathbb{P}(W > 2 \mid W > 1)$. (3 points)

(c) A separate task's duration Y is normal with mean 10 and *variance* 4. Express $\mathbb{P}(8 < Y < 13)$ using Φ. (2 points)

(d) Let U be uniform on $[0, 4]$. Find $\mathbb{P}(U > 3 \mid U > 2)$ and compare it with $\mathbb{P}(U > 1)$. Does U have the memoryless property? (2 points)

5. From a density to a bound

10 points · about 10 minutes

A continuous random variable X has density

$$f(x) = \begin{cases} cx, & 0 < x < 2, \\ 0, & \text{otherwise.} \end{cases}$$

(a) Determine c. (2 points)

(b) Find the CDF $F(x)$ for every real x, and use it to find $\mathbb{P}(1 < X < 2)$. (2 points)

(c) Compute $\mathbb{E}[X]$, $\mathbb{E}[X^2]$, and $\mathrm{Var}(X)$. (4 points)

(d) Use Chebyshev's inequality to give an upper bound for $\mathbb{P}(|X - 4/3| \geq 2/3)$. State why the inequality applies. (2 points)